### 1.6.1 Hemoglobin Buffer system

By the process of metabolism, carbon di oxide is produced in the tissues and enters the blood. This $CO_2$ will be hydrated forming $H_2CO_3$ and this gets ionized yielding $H^+$ and $HCO_3^-$. When the oxygen tension is reduced in the tissues, oxy-hemoglobin will dissociate, thereby producing oxygen and hence reduced hemoglobin is formed.

In lungs, oxy-hemoglobin (strong acid) is formed from reduced hemoglobin releasing hydrogen ions that reacts with bicarbonate and yield carbonic acid. Owing to lower $CO_2$ tension in lung, the shift of equilibrium towards the production of carbon di oxide is enabled which in turn will be continuously eliminated during exhalation

$$H^+ + HCO_3^- \rightleftharpoons H_2CO_3 \rightleftharpoons H_2O + CO_2$$

Reduced hemoglobin will act as anion and accepts the $H^+$ ions to produce acid-reduced hemoglobin (HHb). By the production of weak acid, the arrival of $H^+$ ions are buffered thereby causing little change in pH.

On the return of blood to the lungs, owing to the formation of oxy-hemoglobin (stronger acid ) as mentioned above, these $H^+$ ions are released . Immediately these released $H^+$ ions are neutralized by $HCO_3^-$. This is inevitable for the lungs to release $CO_2$.

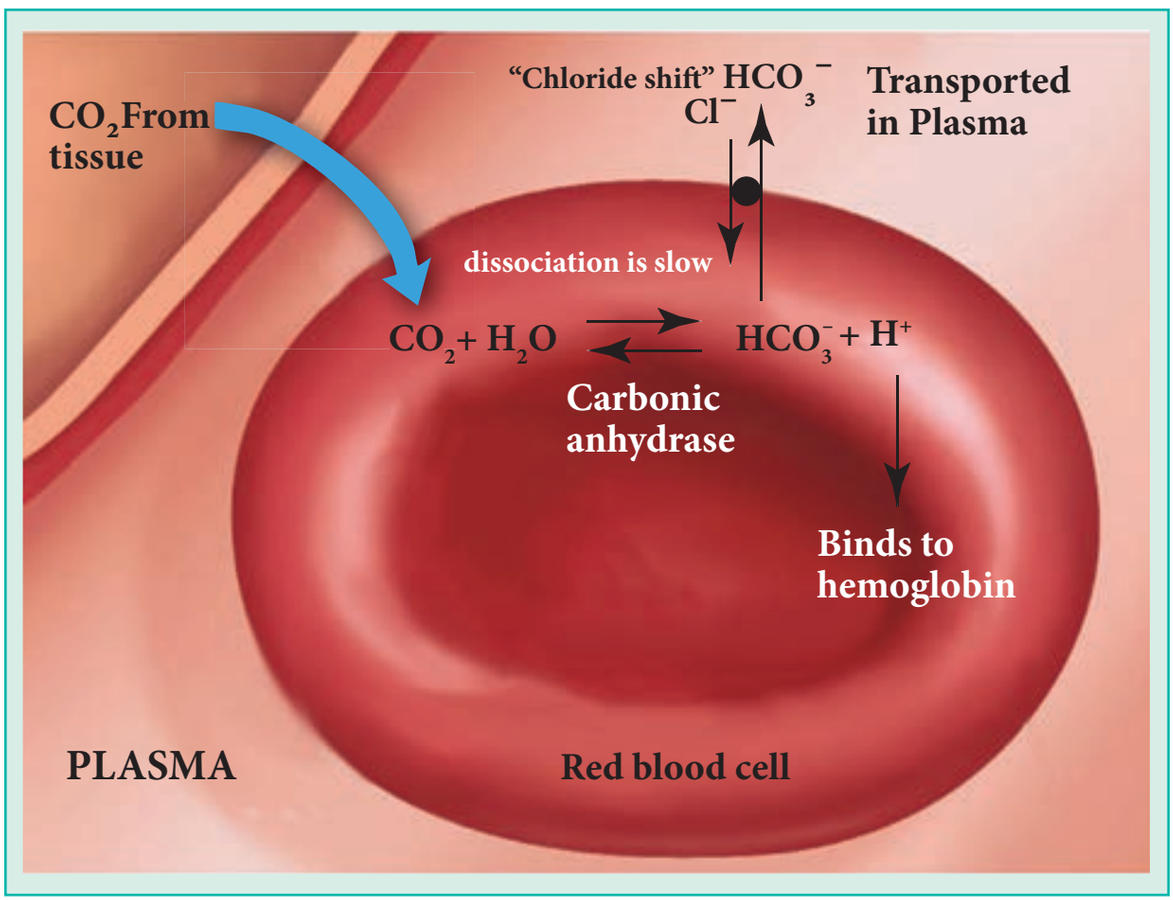



**Figure 9**



XII U1 Cell Membrane.indd 17 4/8/2020 3:03:31 PM